GI COAR

Objectives GI COAR

Clinical Skills

1. Identify important aspects of a HPI relevant to the patient who presents with abdominal pain or gastrointestinal dysfunction.
2. Perform an integrated physical exam of a patient with gastrointestinal complaints, including an abdominal exam and osteopathic structural exam.
3. Create an osteopathic treatment plan for the patient with gastrointestinal complaints.
4. Identify when you would obtain imaging in patients with gastrointestinal complaints.
5. Describe red flags related with gastrointestinal complaints.

Osteopathic Manipulative Medicine

1. Discuss the A (Autonomics), B (Biomechanics), C (Circulation), and S (Screening) aspects of how structural findings could contribute to gastrointestinal pathology.
2. Perform osteopathic manipulative treatment aimed to assist a patient who presents with gastrointestinal disease.
3. Identify how viscerosomatic reflexes may occur due to gastrointestinal disease.

Anatomy

1. Delineate the surfaces of the parietal and visceral peritoneum and their various interconnecting ligaments throughout the peritoneal cavity and explain their sensitivity to pain.
2. Trace the blood supply, innervation, and lymphatic drainage to the stomach.
3. Review the musculature of the posterior abdominal wall including the diaphragm and relate those structures to mechanical respiration and movement of the digestive system.

Biomedical sciences

1. Summarize the classes of agents used to manage GERD.
2. Describe the mechanism of action and major side effects of each drug class.
3. Identify the physiological mechanisms that normally function to defend the esophageal mucosa from damage due to gastroesophageal reflux.
4. Describe the neural and/or hormonal regulation of defense mechanisms protecting the esophageal mucosa from damage due to gastroesophageal reflux.
5. Rationalize the pathophysiological mechanisms by which GERD may develop based on your knowledge of the physiological motility and secretory activities of the upper GI tract

History GI COAR

Location: Outpatient clinic

CC: Heartburn

HPI: 36-year-old male presents with 4-month history of worsening "heartburn." He describes the pain as "in my lower chest" without radiation. The pain began insidiously and has progressively gotten worse. The pain is at least 2/10, at worst 5/10, and currently 3/10. The pain often feels worse after he has eaten and is better taking OTC Tums (he takes 8-10 per day). He has noticed that spicy food really makes the pain worse. It is not better or worse during any time of the day, and he has had occasional episodes of "heartburn" before, but never this bad nor lasting this long. He denies fevers, chills, weight loss, vomiting, diarrhea, sweating or shortness of breath. He admits to intermittent nausea, an occasional dry cough, and episodes of sour tastes in his mouth. He has not recently traveled nor changed his diet.

PMHX: Chronic low back pain since age 27

PSHX: Wisdom teeth removal age 20, appendectomy age 18

Trauma history: 2 motor vehicle collisions-both with concussion, fall off a roof- chronic low back pain

Allergies: No known drug allergies (NKDA), no known food allergies (NKFA), no latex allergies

Medications: Tums OTC (8-10 per day), ibuprofen 800mg tabs 3-6 per day PRN low back pain

Social History: admits to smoking 1 pack per day since age 18, drinks EtOH 2-4 beers per night, denies use of other recreational substances, married, monogamous relationship, 2 children (ages 8 & 12-years-old). Works at an autobody shop, no exercise. Diet consists of coffee for breakfast, fast food for lunch, Monster energy drinks 1-2 per day, and meat/vegetable for dinner.

Family History: Mother -alive 62: HTN, osteoporosis; Father-alive 65: osteoarthritis (knees, back), HTN, MI; Brother-alive 45: HTN, hyperlipidemia; children are healthy

Patient Concerns: "Tums isn't helping anymore- I'm worried that something is wrong."

ROS:

General: denies fevers, chills, night sweats, fatigue, or weight changes

HEENT: denies sore throat, rhinorrhea, congestion, sinusitis, vertigo, jaw pain, bruxism, seasonal allergies

Resp: denies shortness of breath. Admits to "coughing slightly more on most days" (dry cough)

Cardiac: denies chest pain other than as described in HPI, palpitations, shortness of breath with exertion

GI: denies vomiting, hematemesis, diarrhea, constipation, hematochezia, melena, dysphagia of solids or liquids, odynophagia. Admits to epigastric pain, intermittent nausea

Neuro: denies muscle weakness, seizure, loss of consciousness, ataxia, vision changes

MSK: denies arthralgias, myalgias, joint swelling, or joint erythema. Admits to left periscapular thoracic pain, and chronic unchanging low back pain.

Derm: denies rash or skin changes, including jaundice

Endo: denies polyuria, polydipsia; changes in hair, skin, or nails; heat or cold intolerance

Heme: denies easy bleeding or bruising

Physical Exam GI COAR

Vitals: HR: 75, RR 12, BP 130/78, BMI 27, Temp 98.0

General: WD/WN (well developed well nourished) male in NAD (no acute distress), sitting upright.

HEENT: Head NC/AT (normocephalic, atraumatic)

Eyes: No conjunctival injection or icterus

Mouth: Poor dentition with many dental caries, + halitosis, oropharynx clear

Neck: Thyroid without enlargement or nodules, no palpable lymph nodes anterior cervical, posterior cervical, or supra/infra clavicular

Lungs: CTAB (clear to auscultation bilaterally), no R/R/W (rales/rhonchi/wheezes)

Cardiovascular: RRR (regular rate and rhythm), no M/R/G (murmurs/rubs/gallops)

ABDOMEN: normoactive BS x4, soft, slightly tender in the epigastric region, ND (non-distended), no hepatosplenomegaly. No bruit of the aorta or renal vessels.

GU: no CVA tenderness (rectal exam not performed)

Neuro: CN II-XII grossly intact and non-focal

Skin: no rash, no jaundice

<u>Osteopathic Structural Exam</u>:

Head: Hypertonic suboccipital musculature

Cervical: C3 ER_RS_R, C5 ER_LS_L

Thoracic: T4 ER_LS_L, T5-T7 NS_RR_L, T8 ER_LS_L, hypertonic paraspinal muscles worst at T5-T9 on the left, thoracic inlet S_LR_L

Lumbar: Left quadratus lumborum hypertonic, L1 ER_RS_R, L2-L4 NS_RR_L, L5 ER_RS_R

Sacrum: L on R sacral torsion

Pelvis: Right anterior innominate

Ribs: Left rib 12 inhalation somatic dysfunction, left ribs 6-8 exhalation somatic dysfunction

Abdomen: Diaphragm myofascial restriction to the left, myofascial pull to the epigastric region on the left, tenderness at the celiac ganglia, exhalation somatic dysfunction of the stomach

Zink: OA: R; C/T: L; T/L: L; L/S: R

Chapman's point: sternum at the 2nd intercostal space

Clinical Skills GI COAR

Anterior Seated

❏ Observe general posture, distress level, skin (jaundice, pale)

HEENT

❏ Eyes (conjunctiva looking for pallor or jaundice)

❏ Nose (nares patent, mucosa)

❏ Mouth (oropharynx, tonsils looking for lesions, dentition, erosion of teeth)

Neck

❏ Lymphadenopathy (anterior/posterior cervical, supra/infra clavicular)

❏ Auscultation (rate, rhythm, murmurs) **Cardiac**

❏ Auscultation (rales, rhonchi, wheezes, alterations in breath sounds) **Respiratory**

Posterior Seated

Neck

❏ Thyroid

Respiratory

❏ Auscultation (as above)

MSK

❏ Tenderness, asymmetry, alterations in range of motion, tissue texture abnormalities (TART)

❏ Thoracic region T1-T12 (T3-T11 will cover esophagus through small intestine) **video states T5-T12* We are using this chart from ECOP which is the most up to date chart for testing purposes Chart of Visceral Innervation of the Autonomic Nervous System ECOP.pdfDownload Chart of Visceral Innervation of the Autonomic Nervous System ECOP.pdf*

❏ Thoracic inlet (sidebending & rotation)

❏ Ribs (with breath- ribs 1-12)

❏ CVA tenderness

Supine

❏ Abdominal exam (observation, auscultation, percussion, palpation, special tests- Murphy's sign)

❏ Fascial pull

❏ Evaluation of the ganglia (celiac, superior mesenteric, inferior mesenteric)

❏ Diaphragm assessment (sidebending and rotation)

❏ Zink Screen (C-C, C-T, T-L, L-S)

❏ Chapman's points

Differential Diagnosis GI COAR

Differential Diagnosis

Below is a list of potential diagnoses based on the present case. How would you rule in/out some of these differential diagnoses? Use UpToDate or another peer reviewed source to look up each diagnosis and start to create a comparison grid, to understand overlapping symptoms, presentations, physical exam, etc.

Check out UpToDate's Evaluation of the adult with abdominal painLinks to an external site. and Approach to the adult with dyspepsiaLinks to an external site. to aid in your differential diagnoses (DDx).

Gastrointestinal

- Gastroesophageal Reflux Disease (GERD)
- Gastritis/gastropathy
    - NSAID induced
    - Helicobacter pylori infection
    - Alcohol induced
- Peptic Ulcer Disease
- Acute/Chronic pancreatitis
- Esophagitis
    - Erosive
    - Infectious
    - Pill
    - Eosinophilic
- Functional Dyspepsia
- Dysphagia
    - Esophageal motility disorders
    - Rings/webs
- Gastroparesis
- Cholelithiasis
- Biliary Disease
- Inflammatory bowel disease (IBD)

Cardiac

- Acute myocardial infarction
- Atypical Angina
- Microvascular/Vasospastic Angina (Prinzmetal's)
- Pericarditis

Respiratory

- Chronic Bronchitis
- Cough secondary to reflux
- Asthma
- Pneumonia

Musculoskeletal

- Viscerosomatic reflex
- Costochondritis
- Somatic Dysfunction (including, but not limited to)
    - Thoracic region
    - Abdominal/Other region
    - Rib region

When considering the DDx of the patient in this case it is important to consider that epigastric pain could be coming from various systems.

This can include, GI, Cardiac, Respiratory and MSK.

To start breaking down which system is the likely culprit, start with the following questions:

What are two or more factors (from the HPI, ROS, PE) that might make his symptoms seem to be coming from a _cardiac_ source?

- Cough
- Chest pain
- Age
- Smoker
- Family history

What are two or more factors (from the HPI, ROS, PE) that might make his symptoms seem to be coming from a _respiratory_ source?

- Cough
- OMM exam findings (viscerosomatic reflexes)

What are two or more factors (from the HPI, ROS, PE) that might make his symptoms seem to be coming from a _gastrointestinal_ source?

- Epigastric pain
- Pain worse after eating
- Intermittent nausea
- Water brash
- Ingestion of NSAIDS frequently

- Daily intake of caffeine and alcohol
- OMM exam findings (fascial pull, Chapman's points, viscerosomatic reflexes)

Differential DDx Expanded

It is likely that based on our patient's HPI, ROS and PE, that they might have either dyspepsia or GERD. Let's dive deeper into these two diagnoses.

First: What is the difference between dyspepsia and GERD?

<u>Dyspepsia</u> includes one or more of the following: bothersome upper abdominal symptoms often meal related, postprandial fullness (bloating), early satiety, epigastric pain and/or burning, and/or epigastric pain that may or may not be related to meals.

Dyspepsia risk factors:

  - Female sex
  - Younger age
  - Smoking
  - Helicobacter pylori infection
  - NSAID use
  - Acute gastroenteritis
  - Specific psychiatric comorbidities

- <u>Comorbidities:</u>
  - GERD
  - Gastroparesis
  - IBS

75-80% of individuals with dyspepsia have Functional dyspepsia= idiopathic, non-ulcer with no underlying cause on diagnostic exam. *UptoDate*

<u>GERD</u> has a variety of definitions based off of patient-reported symptoms, endoscopy findings, physiologic testing or response to anti-secretory therapy. Clinical symptoms of GERD include heartburn and regurgitation of gastric contents into the mouth or hypopharynx.

Typical symptoms include:

- Heartburn
- Regurgitation
- Hoarseness
- Chronic cough
- Non-cardiac chest pain
- Water brash
- Sore throat
- Dysphagia
- Globus sensations

GERD risk factors include:

- Nicotine use

- Alcohol use
- Coffee
- Obesity
- NSAID use
- Spicy foods
- Fatty foods
- Lying supine soon after eating

Comorbidities:

- Obesity
- Hiatal hernia
- COPD
- Asthma
- Psychological Disorders (bipolar, anxiety, depressive, sleep)

*There is no gold standard for the diagnosis of GERD. Thus, the diagnosis is based on a combination of symptom presentation, endoscopic evaluation of esophageal mucosa, reflux monitoring, and response to therapeutic intervention. (The American Journal of Gastroenterology) ACG Clinical Guideline for the Diagnosis and Management of Gastroesophageal Reflux DiseaseLinks to an external site.

You might hear about a set of diagnostic criteria for functional gastrointestinal disorders called ROME IV. It is important to know that ROME IV criteriaLinks to an external site. exists and what it is. If we look at these criteria:

The ROME IV Criteria for Functional Dyspepsia includes:

Includes one or more of the following:

- Bothersome postprandial fullness
- Bothersome early satiety
- Bothersome epigastric burning
- Bothersome epigastric pain

AND

- No evidence of structural disease at EGD that is likely to explain the symptoms

There is no specific ROME IV criteria for 'GERD', but it has been classified into phenotypes (I've included several articles that speak to defining GERD below)

Esophageal DisordersLinks to an external site.

Critical appraisal of Rome IV criteria: hypersensitive esophagus does belong to gastroesophageal reflux disease spectrumLinks to an external site.

Esophageal Reflux Hypersensitivity: A Comprehensive ReviewLinks to an external site.

After review of both of these topics it is likely that our patient presents with a more prominent GERD diagnosis.

Our patient presents with:

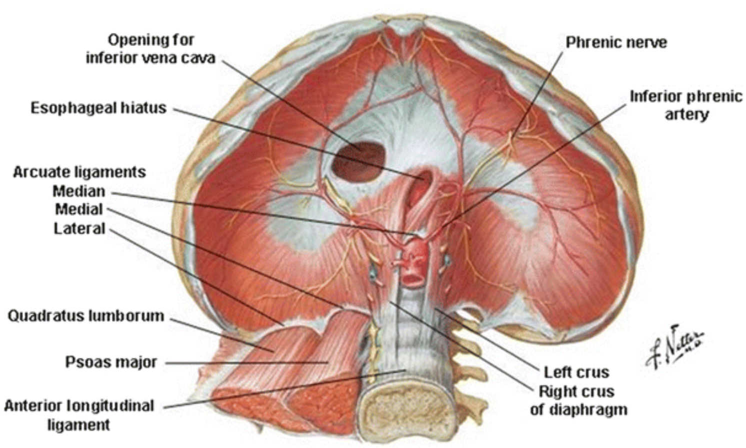
- Heartburn >4 months
- Epigastric pain worse after eating spicy foods
- Partial relief with antacids (Tums)
- Sour taste in mouth
- Chronic dry cough

With the following risk factors:

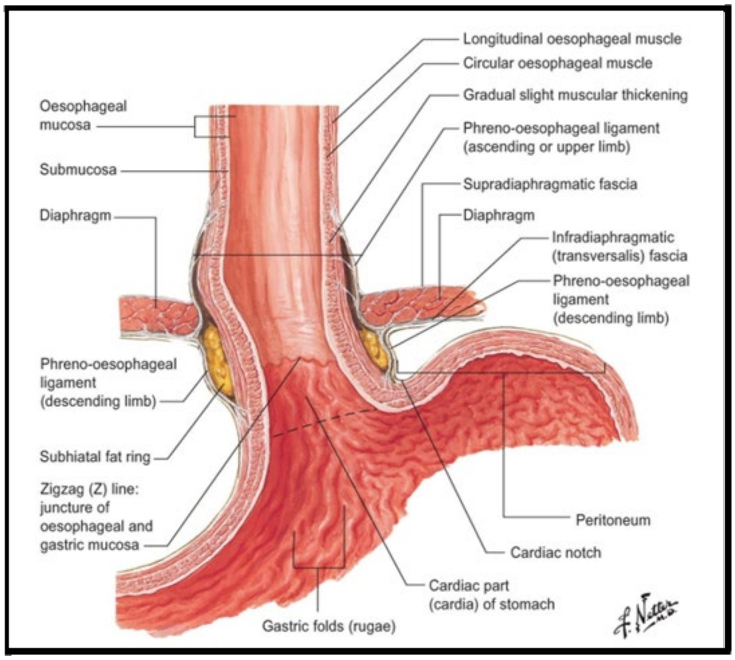
- Smoking
- Alcohol use
- Energy drink intake/high caffeine
- NSAID overuse
- Overweight with BMI 27
- Poor overall diet

Anatomy GI COAR

Diaphragm and Posterior Abdominal Wall



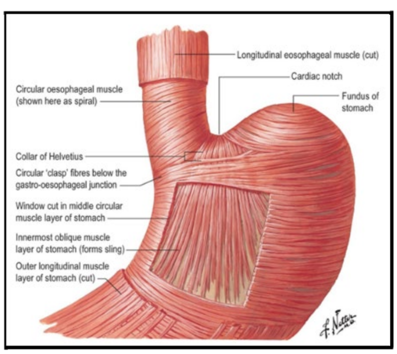
The diaphragm is a continuous sheet of muscle and has attachments to the sternum, ribs, and lumbar spine. The lumbar or posterior portion of the diaphragm attaches to the lumbar spine via the crura (right and left crus). The right and left crura are muscular bands that attach to the anterior longitudinal ligament of the spine, forming the aortic hiatus. The diaphragm also has aponeurotic arches bilaterally, the medial and lateral arcuate ligaments, that provide support and attachment to the spine and ribs. The medial arcuate ligament begins from the transverse process of the L1 vertebra, coursing over the psoas major muscle, and attaching directly to the crus. The lateral arcuate ligament spans from the 12th rib, courses over the quadratus lumborum muscle to the transverse process of the L1 vertebra.



The esophagus passes through the diaphragm at the T10 level along with the anterior and posterior vagal trunks and the left esophageal vessels. It is covered in an adventitial fascial tissue that becomes more organized and becomes the phreno-esophageal ligament that tethers the esophagus to the diaphragm. This ligament has an upper portion and a lower portion attaching the esophagus and stomach to the esophagus.

The diaphragm also has a band of skeletal muscle that attaches to the smooth muscle of the duodenum known as the suspensory ligament of the diaphragm (the Ligament of Treitz). This band is responsible for widening the lumen of the duodenum allowing for food passage. Since it contracts with breath, respiration plays a role in the peristalsis of the duodenum.

Gastroesophageal junction (GE) and lower esophageal sphincter (LES)



Externally, the right side of the esophagus is continuous with the lesser curvature of the stomach and the left side with the greater curvature. Internally, the change between the esophagus is difficult to visualize but can be seen as a zig zag line known as the “Z-line” and represents the change between non-keratinized squamous cell epithelium and columnar epithelium.

The esophageal sphincter is not a specific structure, but a combination of physiological and anatomical features. At the GE junction there are several mechanisms that prevent reflux of gastric contents into the esophagus. One of these is a thickened, tonically contracted band of circular muscle reinforced by the musculature of the right crus of the diaphragm, “clasp-like” fibers on the right side of the stomach, and “sling-like fibers" on the left side. This specialized musculature creates a high-pressure zone creating an air and fluid seal between stomach and esophagus.

Abdominal Viscera and Mesenteries

Peritoneum

The peritoneum is a <u>continuous</u> serous membrane made up of simple squamous epithelium that can be divided into a parietal and visceral layer. Recall that the parietal layer covers the entirety of the abdominal cavity and will reflect from certain areas onto intraperitoneal organs to become the visceral layer. As the peritoneum reflects from the abdominal walls and intraperitoneal organs it becomes a double layer connecting many structures in the abdominal cavity. These structures are classically misidentified as the peritoneal ligaments and contain no connective tissue but are two epithelial layers fused together. These “ligaments” can be divided into a few categories:

<u>Mesentery/mesocolon</u>: Double layer that suspends the gut tube – transverse mesocolon, small bowel mesentery

Transverse mesocolon: connects the transverse colon to the posterior body wall

Small bowel mesentery/root of mesentery: connects the small bowel, jejunum/ileum to the posterior body wall. Both mesenteries contain most branches of the superior mesenteric artery.

<u>Omentum</u>: special type of mesentery that attaches the stomach to other structures – greater omentum, lesser omentum

Greater omentum: attaches from stomach to transverse colon

Lesser omentum: made up of two bands of peritoneum: hepatogastric and hepatoduodenal ligaments. The lesser omentum contains the common bile duct, portal vein, and proper hepatic artery.

<u>Peritoneal ligaments</u>: double layer that attaches more solid viscera to the abdominal wall or other organs

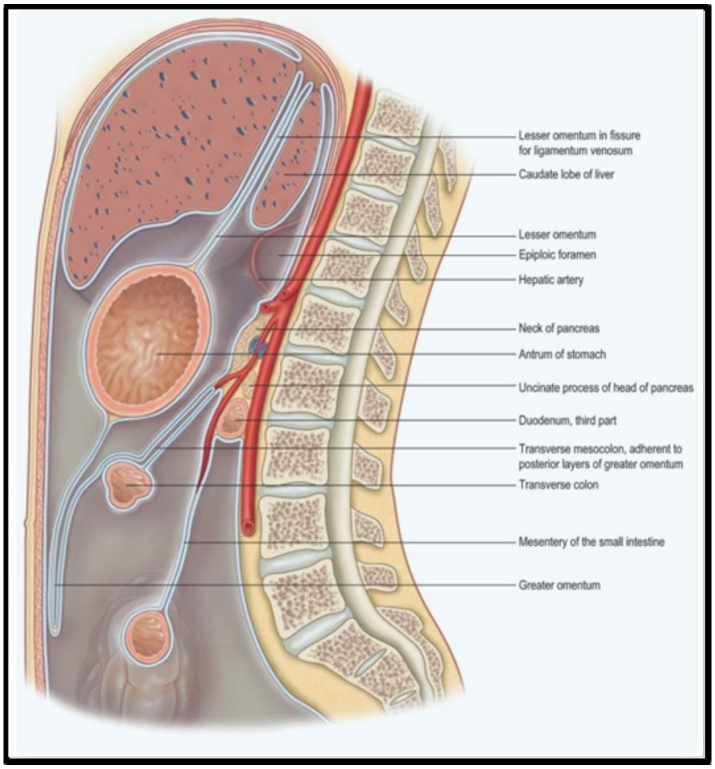
Coronary ligament - circular structure that suspends the liver from the diaphragm

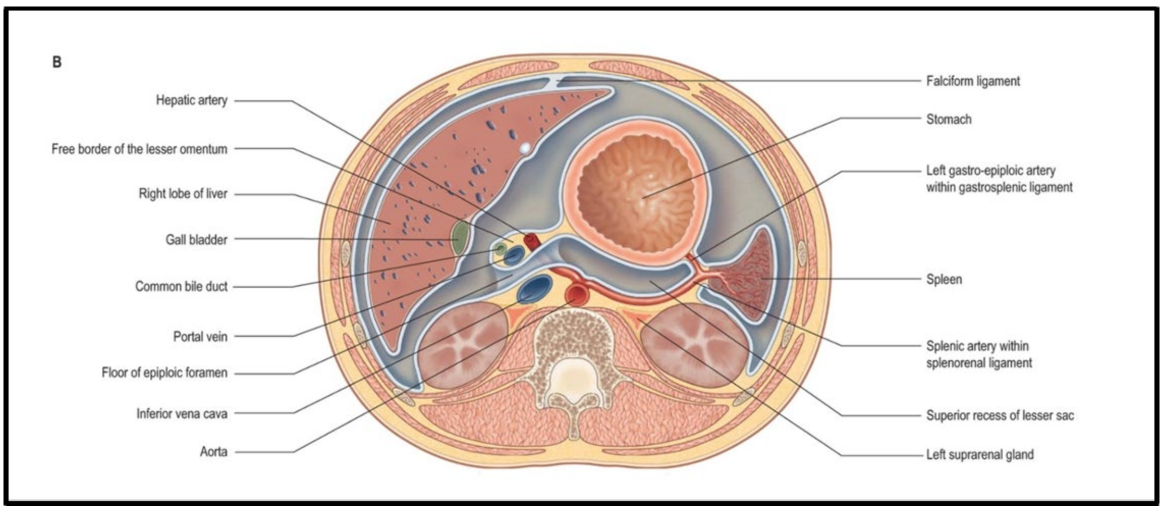
Falciform ligament - anterior continuation of the coronary ligament that anchors the liver to the anterior abdominal wall

Splenorenal ligament: Connects spleen to left kidney - contains part of the splenic artery

Gastrosplenic ligament: Connects stomach to spleen - contains the splenic, short gastric, and proximal left gastro-omental artery.

<u>Peritoneal folds</u>: raised single layer peritoneum coursing over blood vessels or other structures i.e., ureter.





Innervation of the Abdomen

The abdomen is richly innervated by visceral afferent nerves, which are part of the autonomic nervous system. These afferent nerves can be organized into somatic and visceral groups. Somatic afferents refer to the innervation of the abdominal wall and peritoneum which arise from the 6-12 intercostal nerves, ilioinguinal, and iliohypogastric nerves. The visceral nerves are exclusively autonomic and consist of neurons from the sympathetic trunk and the vagus nerve. The sympathetic afferents will convey visceral nociception, while parasympathetic afferents carry neurons that are responsible for hunger, satiation, and nausea.

The innervation to the distal part of the esophagus arises from the esophageal plexus and contains nerves from both the vagus nerve and branches from the sympathetic trunk T(4)5-T9. The rest of the upper GI will receive its innervation from the vagus (CN X) and the greater splanchnic nerve T5-T9.

Parasympathetic: Vagus

Preganglionic neurons - the preganglionic neurons of the vagus nerve exit the medulla oblongata and descend as the right and left vagal nerves. These vagal nerves parallel the esophagus passing through the diaphragm at T10 together. Much of the vagus nerve will enter the celiac and superior mesenteric ganglia and spread throughout the abdominal viscera using the vasculature to find the target organ.

Postganglionic neuron - once the preganglionic neuron reaches the target organ it will synapse on the body wall of that organ in a mural ganglion. Once synapsed the postganglionic neuron will exit to innervate the organ.

Sympathetic:

Preganglionic: Preganglionic sympathetic neurons in the celiac region begin in the lateral horn of the spinal cord at levels T5-T9. The axons then travel through the ventral root, into the spinal nerve, and then follow the ventral rami to the white communicating rami to enter the sympathetic trunk ganglion. Here, the neurons do not synapse but travel medially on individual splanchnic nerves stemming from the T5-T9 ganglia. These splanchnic nerves combine to become the greater splanchnic nerve and pass posterior to the diaphragm where they enter the celiac ganglion and synapse.

Postganglionic: After the synapse onto a postganglionic neuron, the axons adhere to the adventitia of the vasculature and travel to their target organ to perform their sympathetic function.

Upper GI Blood Supply

Most of the upper GI receives its blood supply from the celiac trunk. The celiac trunk exits the abdominal aorta just inferior to the diaphragm and will split into three branches: splenic, common hepatic, and left gastric arteries.

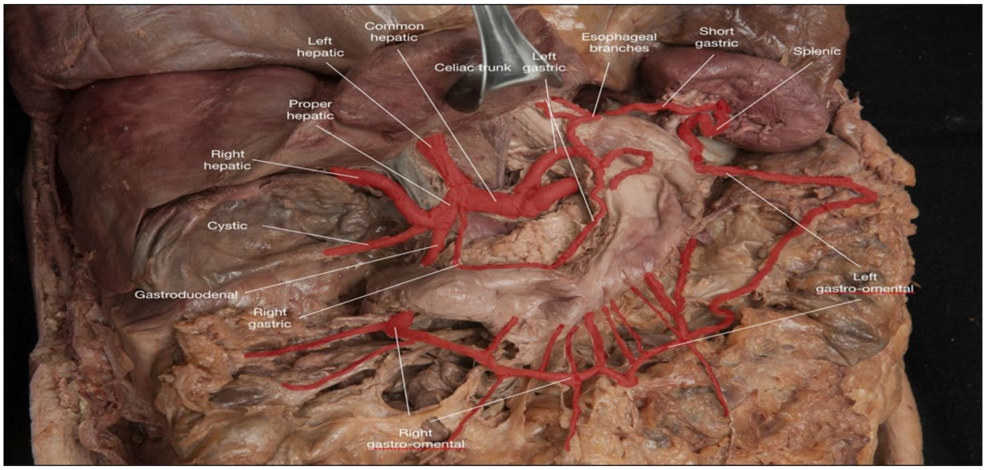
Venous drainage is through many small veins that will either drain to the portal system through the splenic vein or through the azygos system by way of the esophageal veins.

Test yourself: **trace out the blood supply of the celiac trunk using the following terms:**

Celiac trunk	Splenic gastric	Left
Left gastro-omental	Proper hepatic	Gastroduodenal
Right hepatic	Left hepatic gastric	Right

Right gastro-omental Esophageal branches Cystic

Common hepatic Superior pancreaticoduodenal



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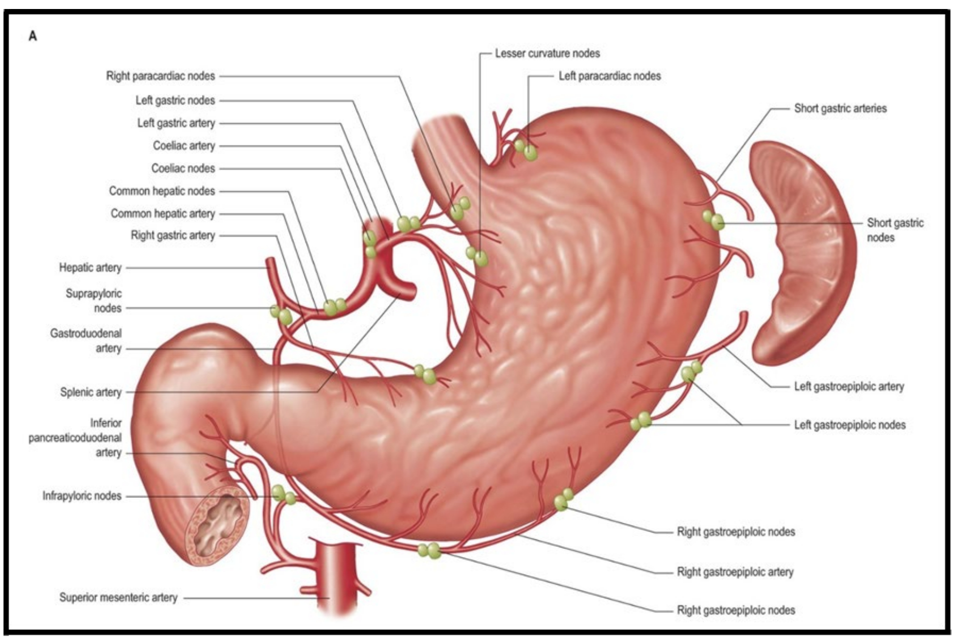
Lymphatic Drainage of the Upper GI tract

The stomach and lower esophagus contain a rich network of lymphatic tissue. At the level of the lower esophagus and upper stomach the drainage flows into the thorax, while the nodes of the lower stomach and pylorus follow the blood supply of the pancreas and duodenum. The diaphragm below shows many named groups of nodes in this area and are important for surgical resection of the stomach and other disease processes. (you will not be tested on the names or locations of the nodes!)



Gastric reflux occurs physiologically but does not usually result in mucosal injury. This is because various gastrointestinal motility and secretory mechanisms protect the esophageal mucosa from damage or minimize the frequency and/or severity of reflux events. The development of GERD fundamentally reflects an imbalance between these defense mechanisms and pathophysiological factors exacerbating gastric reflux episodes. The clinical presentation of the patient in this case study reflects this imbalance.

Eight gastrointestinal motility and secretory mechanisms that defend the esophageal mucosa from damage include:

- <u>Lower esophageal sphincter tone</u>
  - intrinsic tone creates positive pressure gastroesophageal barrier
- <u>Swallowing reflex (primary esophageal peristalsis)</u>
  - primary esophageal peristalsis clears refluxed material
- <u>Secondary esophageal peristalsis</u>
  - periodically clears residual reflux material
- <u>Esophageal mucosa secretory activities</u>
  - mucosal epithelial tight junctions; epithelial membrane H+ extrusion & HCO3- titration
- <u>Salivary secretions</u>
  - acid buffering and dilution actions
- <u>Receptive relaxation</u>
  - stomach region relaxation in response to gastric filling minimizes retrograde gastric pressure
- <u>Gastric migrating motility complex</u>
  - prevents accumulation of acid and cellular debris in stomach during fasting period, which reduces acidity and volume of refluxed material and moderates retrograde pressure
- <u>Gastric acid regulation</u>
  - fasting acid secretion inhibition and postprandial acid secretion regulation moderates acidity of refluxed material

Each defense mechanism is regulated physiologically:

- <u>Lower esophageal sphincter tone</u>- intrinsic myogenic tone with transient inhibitory relaxation via vagus nerve involving neurocrine regulation by VIP/NO mediators
- <u>Swallowing reflex</u>- voluntary motor control in coordination with vagal/glossopharyngeal involuntary neural regulation involving medullary integration
- <u>Secondary esophageal peristalsis</u>- enteric neural control via mucosal mechano-sensory receptors

- Esophageal mucosa secretory activities- epithelial acid secretory channels, Na+/H+ exchanger; Cl-/HCO3- exchanger; activity regulated in response to chemo-sensory receptor to luminal acid levels
- Salivary secretions- dominant parasympathetic autonomic regulation via facial/glossopharyngeal cranial nerves with minor sympathetic regulation via regulation of salivary gland blood flow; cephalic and local mechano- and chemo-sensory receptor regulation
- Receptive Relaxation- regulated in parallel with LES mechanism
- Gastric Migrating Motility Complex- endocrine regulation via motilin secreted from small intestinal APUD cells
- Gastric acid regulation- neural: vagus; endocrine: gastrin; paracrine: histamine/somatostatin multi-level regulation

The pathophysiological circumstances that can cause dysregulation of the mechanisms and increase the risk of developing GERD include:

- Lower esophageal sphincter tone- impairment: excessive transient LES relaxations e.g. belching; anatomic trauma; idiopathically low or loss of intrinsic myogenic tone; secondary factors such as obesity/pregnancy increases retrograde pressure on junction
- Swallowing reflex- impairment: deglutition disorders
- Secondary esophageal peristalsis- impairment: mucosal epithelial damage
- Esophageal mucosa secretory activities- impairment: mucosal epithelial damage
- Salivary secretions- impairment: salivary gland related disorder, Sjogren's; cystic fibrosis; secondary conditions such as dehydration and smoking; medications affecting autonomic function, etc.
- Receptive Relaxation- impairment: parasympathetic disorders; medications affecting autonomic function, etc.
- Gastric Migrating Motility Complex- impairment: small intestinal disorders/resection; gastroparesis; medications and dietary factors affecting gastric emptying and pyloric sphincter tone

Our patient has clinical aspects of his case that are suggestive of pathophysiological factors exacerbating gastric reflux episodes and/or impaired defense mechanisms that contribute to the imbalance causing his GERD. Examples include:

- Chronic back pain: pain response includes heightened sympathetic state, which inhibits GI motility and secretory functions
- Dietary habits: "spicy foods"; high caffeine and EtOH intake, predominant fast-food diet (high fat/sodium) stimulate acid secretion and slow gastric emptying rate
- Smoking: reduces salivary secretions

Eating increases frequency of transient lower esophageal sphincter (LES) relaxation events during swallowing...raising potential for reflux of gastric contents; gastric filling increases retrograde pressure on esophageal-gastric junction; eating stimulates acid secretion, which increases volume and acidity of refluxed material; ingested material composition may slow rate of gastric emptying resulting in increased reflux events...most likely related to patient's fast food diet.

<u>Assessment/Plan GI COAR</u>

While working up your patient- you should note if there are any **Alarm features** (red flags) — Alarm features are suggestive as an indication for direct endoscopy-concern for gastrointestinal malignancy-include:

- New onset of dyspepsia in patient ≥60 years
- Evidence of gastrointestinal bleeding (hematemesis, melena, hematochezia, occult blood in stool)
- Iron deficiency anemia
- Anorexia
- Unexplained weight loss
- Dysphagia (difficulty swallowing)
- Odynophagia (painful swallowing)
- Persistent vomiting
- Gastrointestinal cancer in a first-degree relative

Assessment: Based on our HPI, ROS, PE and review of our Differential Diagnoses our patient is likely presenting with **<u>GERD</u>**.

Our patient presents with:

- Heartburn >4 months
- Epigastric pain worse after eating spicy foods
- Partial relief with antacids (Tums)
- Sour taste in mouth
- Chronic dry cough

With the following risk factors:

- Smoking
- Alcohol use
- Energy drink intake/high caffeine
- NSAID overuse
- Overweight with BMI 27
- Poor overall diet

Plan:

1. Rule in/out Alarm symptoms and screening for Barrett's esophagus

Per UpToDate:

Clinical features

Alarm symptoms suggestive of gastrointestinal malignancy

- New-onset dyspepsia in patient ≥60 years old
- Evidence of gastrointestinal bleeding (hematemesis, melena, hematochezia, occult blood in stool)
- Iron deficiency anemia
- Anorexia
- Unexplained weight loss
- Dysphagia
- Odynophagia
- Persistent vomiting
- Gastrointestinal cancer in a first-degree relative

Screening for Barrett's esophagus*

Chronic GERD (at least 5 years of persistent symptoms) plus 3 of the following additional risk factors for Barrett's esophagus*:

- Age >50 years
- Male sex
- Non-Hispanic white individuals
- Obesity
- Tobacco use (past or current)
- First-degree relative with Barrett's esophagus and/or esophageal adenocarcinoma

Luminal abnormality on abdominal imaging of upper gastrointestinal tract

Upper endoscopy is indicated in individuals with these features. However, most individuals with typical reflux-like symptoms of heartburn or regurgitation do not need initial endoscopy.

Our patient did not exhibit any of these features. If he had, we would have wanted to get a Gastrointestinal (GI) Referral for esophagogastroduodenoscopy (EGD).

2. **Treatment**: For patients who do not need to progress to EGD as initial management, treatment can be based on symptoms severity, frequency and timing.

<u>Symptoms Frequency</u>:

- Intermittent: <2 episodes per week
- Frequent: 2 or more episodes per week

<u>Symptom severity</u> is considered mild or moderate/severe based on how their symptoms impair quality of life

GERD is often a multifactorial process and education, and counseling should be had and should include the potential underlying cause (pathophysiology), role of self-management, and lifestyle modifications to control symptoms.

Lifestyle changes for initial management of mild acid reflux could include:

- Discussion on weight loss
- Elevating the head of the bed (6-8 inches) may be helpful to some patients (e.g., overweight or night heartburn)
- Avoid acid reflux-inducing food (e.g., excessive caffeine, chocolate, alcohol, peppermint, fatty foods). Dietary change may not be helpful to all patients. See below triggers
- Chew gum or use oral lozenges to increase saliva production. Saliva neutralizes refluxed acid
- Quit smoking (smoking reduces saliva in the mouth and throat)
- Avoid late meals
- Avoid tight fitting clothing

Avoid Common triggers and modulators of reflux-like symptoms[*]

Category	Trigger or modulator
Eating behaviors	• Eating too fast • Large portions • Eating past satiety • Eating too close to bedtime
Dietary factors	• Fatty/fried foods • Spicy foods (capsaicin) • Excess alcohol • Excess coffee

	• Carbonated beverages • Patient-specific trigger foods
Lifestyle	• Weight gain • Tight garments • Belts • Tobacco use • Excessive exercise
Medications	• Anti-inflammatory medicines • Antihypertensives • Erectile dysfunction medicines
Emotional or behavioral factors	• High-stress environment • Hypervigilance

* In most instances, only some of these factors are pertinent to an individual patient. Clinicians should assess for specific triggers and individualize recommendations to avoid guidance

that is overly restrictive or overwhelming.

Make an Individualized Plan

Consider use of an individualized plan as noted below from UpToDate.

Individualized plan for lifestyle changes to manage reflux-like symptoms

Healthy eating	**Night-time behaviors**	**Exercise**	**Medications**

Add in fruits or vegetables at each meal (avoid citrus).	Finish eating approximately 3 hours before lying down.	Accumulate 20 to 30 minutes of physical activity on most days of the week such as walking, swimming, dancing, exercise classes, or cleaning.	__________________
Use plant-based fats over animal fats.	Wear loose clothing to reduce pressure around the belly.	Add in 2 days of strength and flexibility training such as weight training, yoga, Pilates, etc.	__________________
Opt for whole grains such as whole wheat, brown rice, oats, teff, millet, quinoa.	Practice deep breathing or other stress reduction techniques before sleep.	Incorporate activity into lifestyle. If you track steps, aim for >7000 to 10,000 steps on most days (5 to 8 km).	__________________
Swap out animal proteins for plant-based options, eg, lentils, pulses, seeds, nuts, and legumes.	Avoid alcohol before bed.		__________________
Serve smaller portion sizes to help you reduce meal volume.	Elevate head of bed when sleeping, ideally using a wedge pillow or by adjusting mattress or head of bed.		__________________
Use smaller plates and utensils to feel satisfied with smaller amounts.	Lie on left side to minimize reflux.		
Choose water or tea over high-sugar drinks.			
Eliminate carbonated beverages and caffeine if they trigger symptoms.			
Enjoy small desserts a few days in a week or substitute with fruit to finish a meal.			
Limit alcohol.			
Schedule meals to avoid grazing.			

Behavior Changes: Focus on lifestyle changes also include behavioral changes such as:

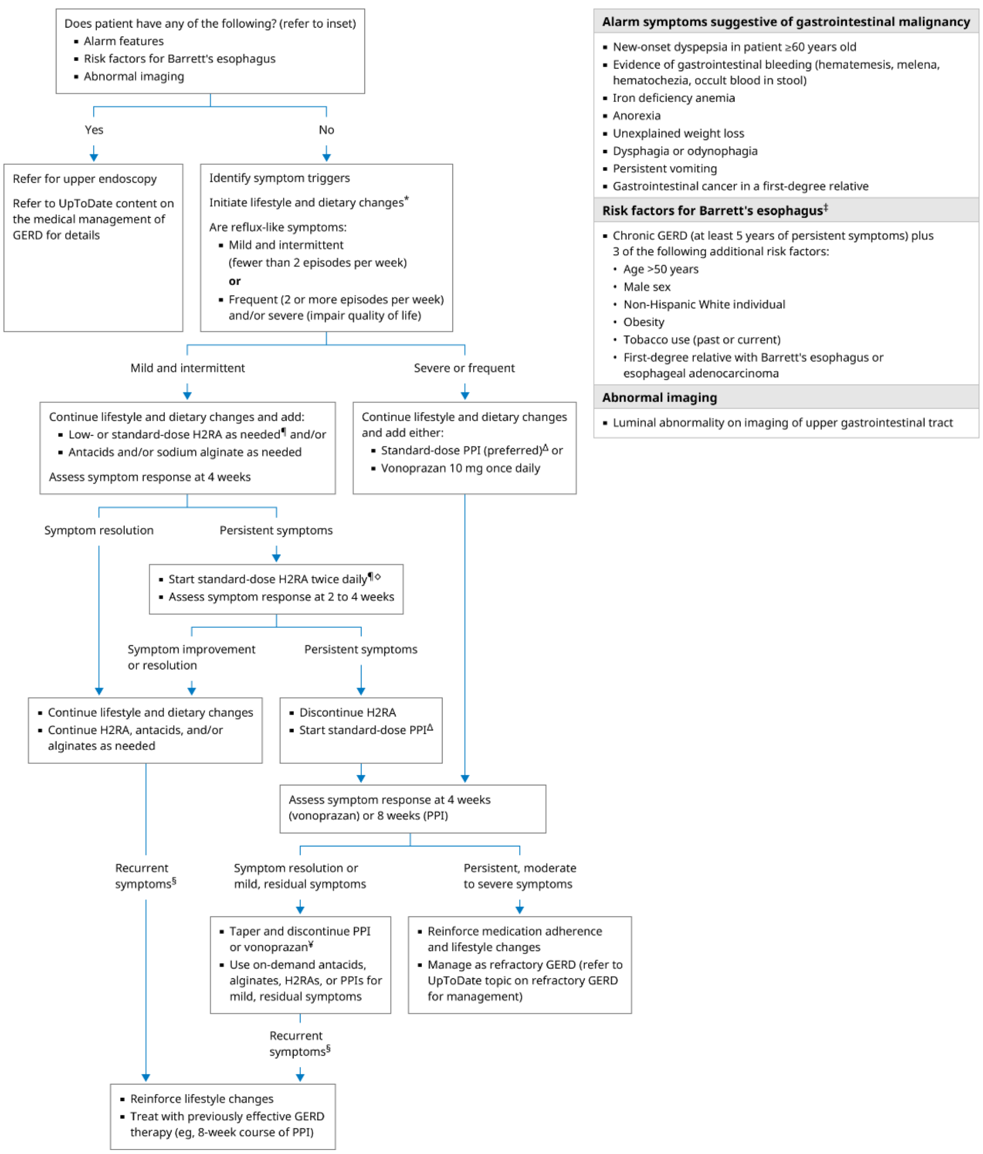
- Avoiding late meals
    - 3 hrs between a meal and lying down for bed
- Sleeping in the left lateral decubitus position
    - gravity keeps gastric contents away from the gastroesophageal junction as the stomach now lies below the esophageal junction
- elevating the head of the bed

Have discussions around weight loss, smoking cessation, avoidance of triggers as noted above, including alcohol.

Consider work on abdominal breathing exercises, referral to nutritionist, OMM.

Initial Pharmacotherapy

Initial empiric management of adults with reflux-like symptoms



Antacids are used for short term relief of acid reflux - e.g., Tums, Mylanta

1. MOA: They act by neutralizing stomach acid – do not prevent GERD so best for relief of MILD GERD symptoms.
2. Major side effects include belching/bloating (bicarbonate), constipation (Al), diarrhea (Mg).
3. Excessive use can cause calcium-related derangements

Histamine antagonist (H2 blockers) - e.g., Cimetidine (Tagamet)

1. They reduce production of stomach acid by inhibiting H_2 receptor (a Gs-coupled receptor) activity, leading to a decrease in cAMP pathway and a decrease in proton pump translocation to the parietal cell apical membrane. They are less effective than PPI for continuous effect.
2. Major side effects for cimetidine include potential drug interactions (mostly CYP-based), gynecomastia (increased prolactin), galactorrhea (increased prolactin).
3. Note rare but severe SE of mental status changes, esp. in special populations: increased risk in elderly or renal/hepatic dysfunction
4. Of note, they are metabolized by cytochrome P450 in the live and cimetidine is the most potent inhibitor of the P450 system and can lead to significant drug interactions including amiodarone, chloroquine, hydroxychloroquine, glipizide, itraconazole, and many others.

For Moderate to severe symptoms consider Proton pump inhibitors (PPI)

Proton pump inhibitors (PPI)-e.g., Omeprazole

1. MOA: They act by irreversible binding to H+/K+ -ATPase pump, which mediates H^+ release to the gastric lumen. Often recommended for trial of 6-8 wks.
2. Major side effects include GI infections (e.g., C. diff due to altered gut flora), reduced drug absorption (i.e., acid-dependent drugs), and reduced B12 absorption.

If symptoms are not relieved, what would you recommend next?

- Confirm PPI being taken on an empty stomach (food decreases bioavailability)
- An alternative PPI may be used
  - Rationale: for "fast metabolizers" (CYP2C19 variant), omeprazole may be metabolized too quickly to reach therapeutic concentrations; consider e.g., esomeprazole or rabeprazole, which are more independent of CYP2C19 activity
- Dosage or frequency of PPI may be increased
- Trial of Vonoprazan – see below

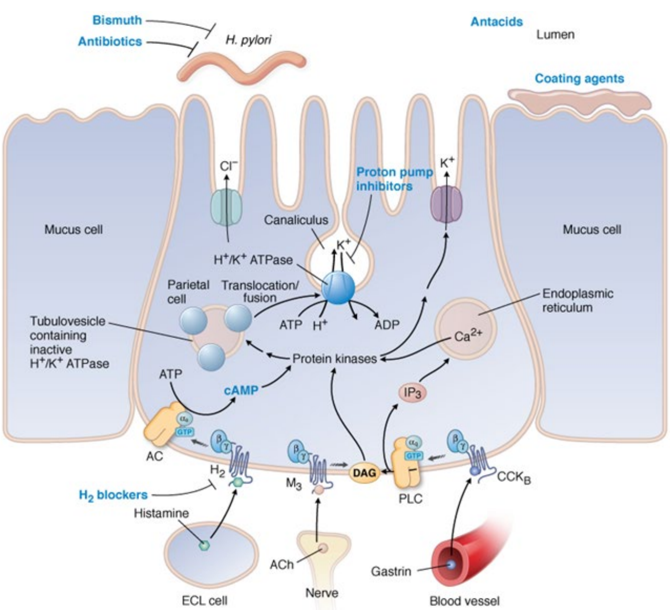
- Consider testing for H. pylori and treating if confirmed.
- Further testing to confirm the diagnosis
- Surgical procedure may be considered
- For more, see e.g., PPI Refractory Gastroesophageal Reflux DiseaseLinks to an external site.

****Recent update in GERD management****

From UpToDate

- Vonoprazan for nonerosive gastroesophageal reflux disease (August 2024)

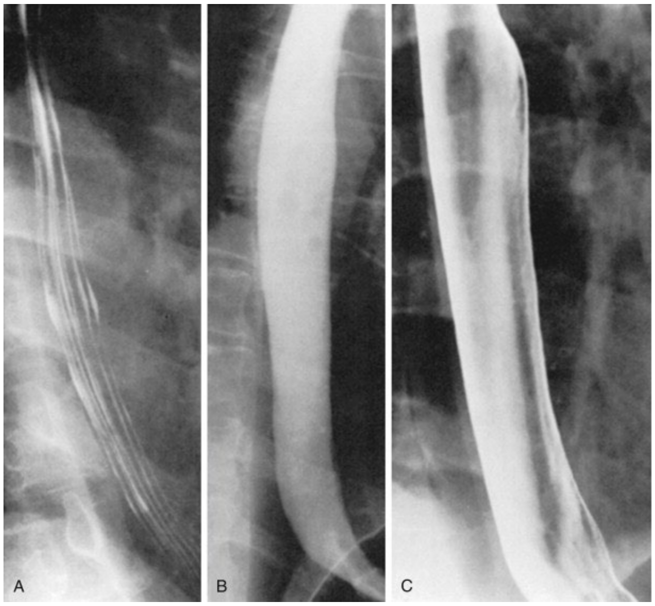
"VonoprazanLinks to an external site., a potassium-competitive acid blocker, rapidly heals erosive esophagitis in patients with gastroesophageal reflux disease (GERD), but its effectiveness for treating nonerosive GERD has been uncertain. In a randomized trial in 772 adults with frequent reflux-like symptoms and normal upper endoscopy, vonoprazan 10 and 20 mg daily reduced heartburn symptoms more effectively than placebo (45 and 44 percent versus 28 percent heartburn-free days, respectively) [1Links to an external site.]. A trial that compared on-demand vonoprazan with placebo reported similar findings [2Links to an external site.]. Based on these results, the US Food and Drug Administration approved vonoprazan for the treatment of nonerosive GERD in July 2024. However, because vonoprazan is expensive and has not been directly compared with proton pump inhibitors (PPIs) in this setting, we continue to suggest PPIs for initial management of nonerosive GERD."



Radiology GI COAR

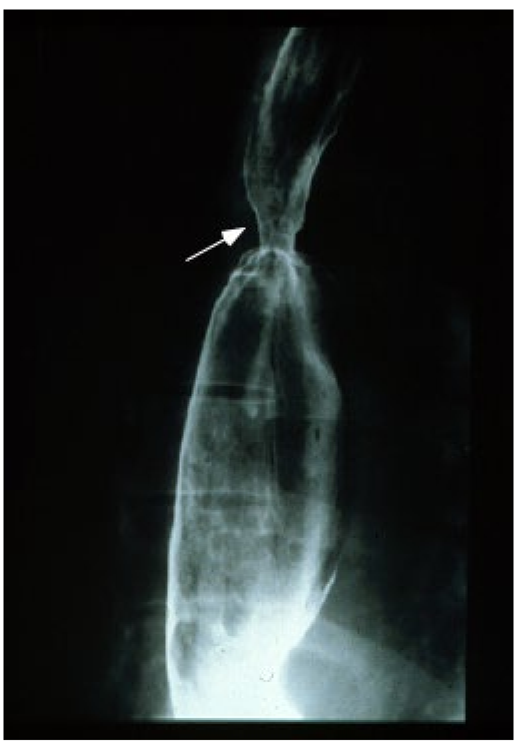
There are limited radiographic exams associated with GERD

Double contrast barium swallow studies are limited in the diagnosis of GERD. Example:



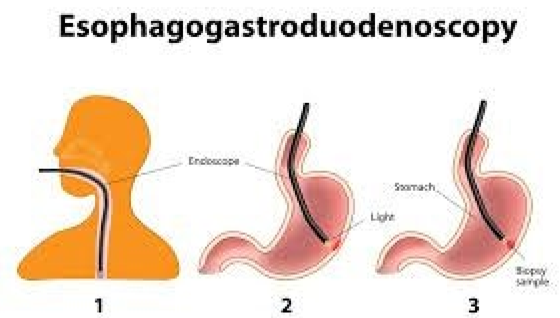
*Radiology Key

This type of study is more sensitive in detecting stricture and characterizing morphology of hiatal hernias.

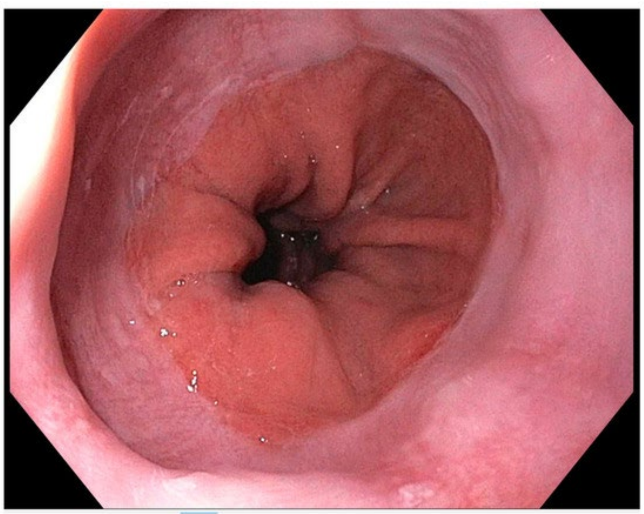


Stricture *UpToDate

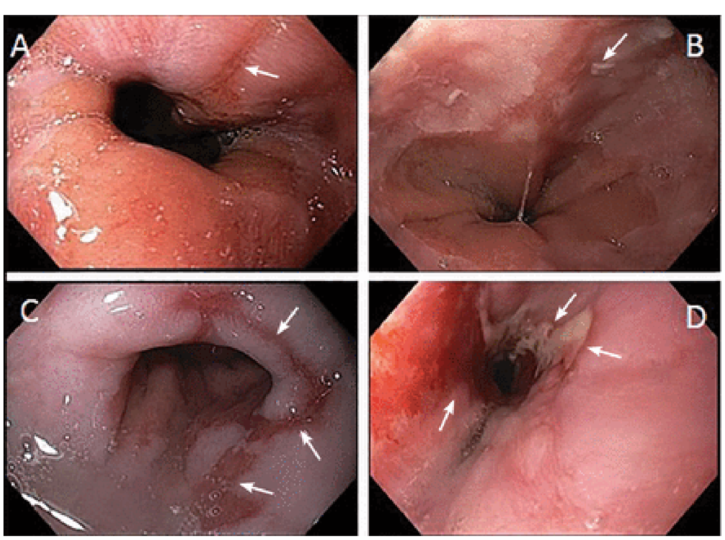
Often an Esophagogastroduodenoscopy (EGD) is used to further evaluate



*Haymarket



Endoscopic image of the gastroesophageal junction (GEJ) using the high-definition (HD) endoscopy. *[Endoscopic Advances in the Diagnosis and Management of Gastroesophageal Reflux DiseaseLinks to an external site.]



Endoscopic views of esophagitis grades. (A) Grade A—1 or more mucosal breaks (arrow) no longer than 5 mm that do not extend between the tops of two mucosal folds. (B) Grade B—1 or more mucosal breaks (arrow) longer than 5 mm that do not extend between the tops of two mucosal folds. (C) Grade C—1 or more mucosal breaks (arrows) that are continuous between the tops of 2 or more mucosal folds, but involve less than 75% of the circumference. (D) Grade D—1 or more mucosal breaks (arrows) that involve at least 75% of the esophageal circumference. *GERD: A practical approachLinks to an external site.

Osteopathic Considerations (ABCS) GI COAR

Remember that we treat *people* as individuals, NOT clinical conditions. Therefore, there is no specific technique(s) for a specific diagnosis including epigastric pain. Each patient, regardless of the clinical diagnosis, requires a structural diagnosis to determine what treatment best suits their clinical presentation.
There is no algorithm for the osteopathic treatment of patients since no one person will have the same constellation of musculoskeletal abnormalities. We can, however, consider each patient in the context of the A (Autonomics), B (Biomechanics), C (Circulation), S (Screening) as a way to begin to discover where we may want to aim our diagnosis and treatment.

Optional Reading: Foundations of Osteopathic Medicine, 4th edition, Chapter 28, Part I. Abdominal Region pg 689-695

Treatment Goals of OMT in a patient with abdominal complaints

- Address viscerosomatic reflexes (TART reflecting homeostatic disturbances)
- Decrease pain
- Decrease segmental facilitation (somatovisceral responses)
- Enhance circulation
- Improve organ function
- Decrease allostatic load promoting better health

A (Autonomics)
Parasympathetics
■ The vagus nerve supplies the parasympathetic control to the GI tract all the way to the transverse colon. Treatment of the OA region may be helpful in addressing parasympathetic tone.
Sympathetics

- T3-6-Esophagus
- T5-T9- Stomach, Liver, Gallbladder, Spleen, Pancreas, Duodenum
- Increased sympathetic tone to the esophagus and stomach causes an inhibitory effect on GI muscles and an inhibitory influence over mucosal secretions
    - Blood flow regulation occurs via vasoconstriction

B (Biomechanics)
Think about any biomechanical restriction around the organ. Be sure to check for fascial restriction of the:
■ Stomach
■ Esophagus

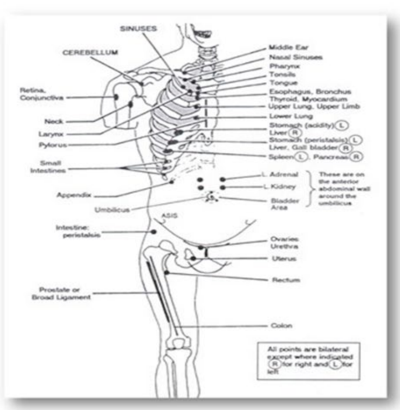
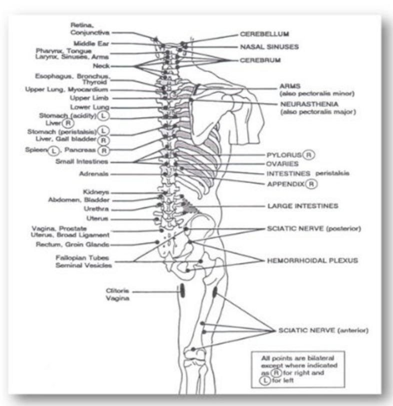
■ Thoracic diaphragm (through which the esophagus must travel)

- Anterior aspect at the costal margins
- Rib 6-12 motion
- 12th rib and posterior attachments of the diaphragm (arcuate ligaments and crura- over the iliopsoas and quadratus lumborum)
- Attachments at L1-L3

Consideration for any abdominal surgical scars (fascial restriction)
Also assess for fascial pull/drag (may be completed prior to conducting your abdominal palpatory exam)
C (Circulation)
Consider adequate vascular/lymphatic flow to the stomach and esophagus, which may be treated by addressing the
■ Diaphragm
■ Thoracic inlet
S (Screening)
Don't forget to look at your patient as a whole
■ History of trauma- can impact any of the ABC's leading to symptoms
■ Zink screen- fascial restriction can lead to poor lymphatic/vascular movement and potentially limit biomechanical motion of organs
■ Mind, body, spirit connection (life stressors, mood). Stress and mood can play a big role in GERD, IBS, or other GI symptoms.
■ Behavioral contributions (diet, lifestyle, exercise, sleep, drugs/EtOH) as lifestyle choices can have a big impact on symptoms

Osteopathic Considerations Treatment GI COAR

Chapman Points

Esophagus:

- Anterior: 2nd rib interspace
- Posterior: between the SP and TP of T2

Upper lung

- Anterior: 3rd rib interspace
- Posterior: Between T3/4 TPs and B/W SPs and TPs of T3/T4

Lower lung

- Anterior: 4th rib interspace
- Posterior: between SPs and TPS of T4-T5

Heart

- Anterior: 2nd rib interspace
- Posterior: between SP and TP of T3

Small intestine

- Anterior: intercostal spaces at costochondral junction b/w ribs 8-9, 9-10, 10-11
- Posterior: between SP and TP of T8, T9, T10

Large intestine

- Anterior: mid-thigh in the TIB from greater trochanter to just above the knee
- Posterior: triangular region with corners at TP of L2, TP of L4 and iliac crest

Appendix

- Anterior: tip of the 12th rib
- Posterior: at the TP of T11

Pancreas

- Anterior: right 7th rib interspace
- Posterior: between TP of T7-T8; between SP and TP of T7-T8 on the right

Liver/gallbladder

- Anterior: right 5th/6th rib interspace
- Posterior: between the SP and TP of T6-T7 on the right

Fascial Pull\Fascial Drag

After abdominal auscultation and prior to superficial and deep palpation, place your hand gently on the abdomen, fingers pointing cephalad. Feel where your hand is pulled to:

- Superior or inferior
- Left or right
- Pulling toward any particular organ

Can induce motion in superior/inferior/left/right/CW/CCW motions

Paraspinal Inhibition

You may treat the viscerosomatic reflexes related to GERD/esophageal and stomach disorders via paraspinal inhibition

Possible Clinical Indications: back pain, GERD, ileus, viscersomatic reflexes

Possible TART Findings/Diagnosis: Rt/Lt/Bilateral hypertonicity of paraspinal muscles in the thoracic or lumbar area

Example Diagnosis: Left lumbar paraspinal muscle hypertonicity

Patient Position: Supine

Physician Position: Seated on either side of the table

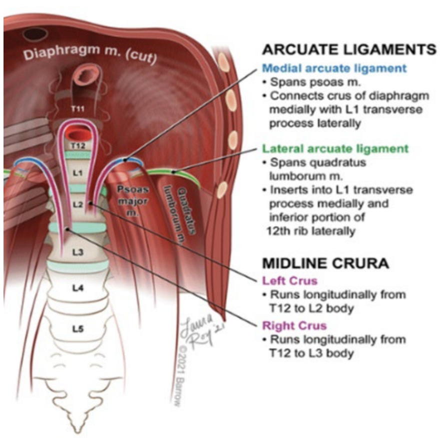
Treatment:

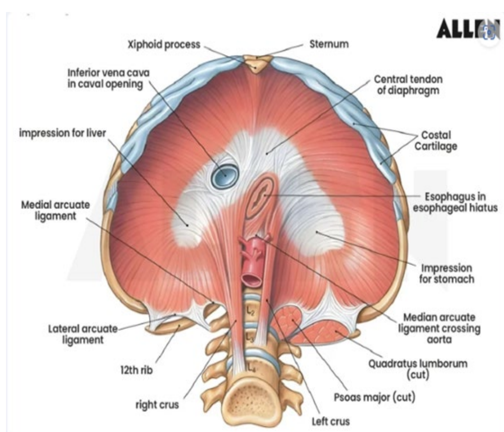
1. With both palms up, the physician's hands are placed so that the patient's spinous processes are between the physician's fingertips and hypothenar/thenar eminences

2. With anterior pressure, using a fulcrum by pressing against the table through the forearms, inhibition is induced through the paraspinal tissues. The pressure can be increased by increasing the tension in the hand/fist.
3. Fingertips and thenar eminences are brought together. (If hands are too small, the physician can inhibit on the ipsilateral side or perform anterior inhibitory pressure with increasing tension through the hand.)
4. The physician holds the inhibitory pressure until a change in tissue tensions is appreciated.
5. Reassess

BLT Release of Lower Esophageal Sphincter/Crus of the Diaphragm
Possible Clinical Indications: GERD/Esophageal/Stomach disorders
Possible TART Findings/Diagnosis: Rt/Lt/Bilateral paraspinal hypertonicity, Rt/Lt/Bilateral 12th Rib inhaled/exhaled
Example Diagnosis: Rt Rib 12 inhaled
Patient Position: Supine
Physician Position: Seated at the patient's Rt side (ipsilateral to the dysfunction)
Treatment:

1. The physician places one hand under the patient's back to establish contact with the right 12th rib (contact/palpation hand)
2. The physician's other hand is placed beneath the contact hand (action hand). The physician now assesses motion at the right 12th rib by lifting anteriorly and providing a lateral traction at a vector between 90◦ from the spine towards the right ASIS along the long axis of the rib
3. The physician then established a balanced ligamentous tension (BLT) between the articulation of the rib and the corresponding vertebra, arcuate ligaments and the right/left crus of the diaphragm
4. The physician fine tunes to maintain the balanced tension and monitors until a release is felt
5. Reassess

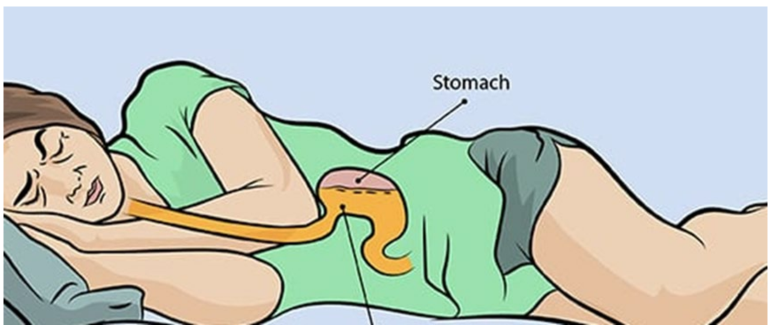




Still Technique for the Esophagus – Direct – Long Lever
Possible Clinical Indications: GERD/Esophageal/Stomach disorders
Possible TART Findings/Diagnosis: Inhaled Stomach, fascial pull/drag restriction
Example Diagnosis: Inhaled Stomach, Esophageal restriction

Patient Position: Right lateral recumbent
Physician Position: Standing behind the patient
Treatment:

1. The physician contacts the lesser curvature of the stomach, palpating for superior tension on the esophagus via this contact
2. The physician then provides a traction inferiorly and laterally to engage the esophagus (tension is felt cephalad from the esophagus)
3. The physician takes up any slack in the tissues with each exhalation breath
4. The physician progressively increases tension from above by having the patient slowly extend their head and neck
5. The physician continues to monitor until they can feel a release (change in the cephalad tension on the esophagus)
6. Reassess



Osteopathic Research GI COAR

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